Unit 3.1 Student Notebook Printout

Print a day at a time, one copy per student. Cut along each title bar and glue where the notebook setup slide shows the box.

Unit pages

U-1 Unit CER page (glue in Day 1, add a row every day)

New page

Question: Why is the seafloor so much younger than the continents?

My first claim (Day 1):

Day	Evidence (use a number)	What it shows

My claim after Day 6:

U-2 Frayer card sheet (homework, finish by Day 7)

Keep in notebook

For each word write a short definition in your own words, draw a quick picture, and give one example.

Word	Definition in my own words	Picture	Example
lithosphere (plate)			
plate boundary (divergent, convergent, transform)			
mid-ocean ridge			
subduction			
trench			
magnetic reversal			
craton (a continent's ancient core)			
impact crater			

Day 1: Where Did the Old Ocean Floor Go?

D1-D Hook data

Left page, above I notice and I wonder

Place on the ocean floor	Age of the rock
Right at the Mid-Atlantic Ridge	0 (forming now)
Middle of the Atlantic, about 2,000 km from the ridge	about 80 million years
Atlantic seafloor next to the east coast of North America	about 180 million years
Right at the East Pacific Rise	0 (forming now)
Western Pacific, east of Japan (the oldest on Earth)	about 180 million years

Continental material	Where	Age
Oldest mineral (a zircon crystal)	Jack Hills, Western Australia	about 4.4 billion years
Oldest known rock (Acasta Gneiss)	Northwest Territories, Canada	about 4.0 billion years
Very old rock (Isua belt)	Greenland	about 3.8 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years

D1-A Site list for the plotted map

Left page, beside your world map

Plot each site with a **red dot** for a volcano and a **blue X** for an earthquake. Each partner plots five.

Site	Type	Latitude	Longitude
Mount St. Helens, Washington, USA	volcano	46.2° N	122.2° W
Mount Fuji, Japan	volcano	35.4° N	138.7° E
Popocatépetl, Mexico	volcano	19.0° N	98.6° W
Hekla, Iceland	volcano	64.0° N	19.7° W
Erta Ale, Ethiopia	volcano	13.6° N	40.7° E
Tohoku, Japan (2011, magnitude 9.1)	earthquake	38.3° N	142.4° E
Maule, Chile (2010, magnitude 8.8)	earthquake	35.9° S	72.7° W
Haiti (2010, magnitude 7.0)	earthquake	18.4° N	72.5° W
Turkey (2023, magnitude 7.8)	earthquake	37.2° N	37.0° E
Kīlauea, Hawai‘i, USA	volcano	19.4° N	155.3° W

The points make _____ . They match the edges of the _____ .

D1-B Plate boundary table

Right page, below your clay-model sketches

Boundary	What the plates did	What formed	Real example
Divergent			
Convergent			
Transform			

Where is ocean crust made? _____ Where is it destroyed? _____

Day 2: Reading Rock Ages

D2-A Seafloor ages and color key

Left page

Distance from ridge (km)	Crust age (million years)
0	0
500	20
1,000	40
1,500	60
2,000	80
2,500	100
3,000	120

Age (million years)	Color
0	red
20	orange
40	yellow
60	green
80	blue
100	purple
120	dark purple

D2-B Seafloor strip: color each box using D2-A

Left page

3,000 km	2,500	2,000	1,500	1,000	500	RIDGE	500	1,000	1,500	2,000	2,500	3,000 km

As you move away from the ridge, the crust gets _____ .

D2-C Graph grid: age vs. distance

Right page

Mark a dot in the box where the distance column meets the age row. Connect the dots.

Age (million yrs)	0 km	500	1,000	1,500	2,000	2,500	3,000 km
120							
100							
80							
60							
40							
20							
0							

Rate = distance ÷ time = _____ km per million years = _____ cm per year. Age at 3,500 km: _____ million years.

D2-D North America crust ages and strip

Right page

Region of North America	Direction and distance from the central core (km)	Age of the crust
Pacific Coast Ranges	west, about 2,800	about 50 million years
Rocky Mountain region	west, about 1,800	about 150 million years
Canadian Shield (the central core)	0	about 2.7 billion years
Grenville Province (Great Lakes to Quebec)	east, about 1,200	about 1.1 billion years
Appalachian Mountains	east, about 2,000	about 400 million years

Color the oldest region dark brown and the youngest light yellow. Choose three colors in between.

Pacific Coast Ranges (west)	Rocky Mountains	Canadian Shield (core)	Grenville Province	Appalachians (east)

Day 3: Reading Deep Time and Early Earth

D3-A Crater counts (from the crater-counter simulation)

Left page

Time step	Moon craters	Earth craters
1 billion years		
2 billion years		
3 billion years		
4 billion years		

Which surface looks older? How can you tell?

Why does Earth show so few craters?

A surface has a lot of craters. Is it old or young? Why?

D3-B Timeline table and old materials

Right page, with your 45 cm strip

Scale: 1 cm = 100 million years. Position = (4,500 − age in millions of years) ÷ 100.

Event	Age (millions of years ago)	Position on the 45 cm strip (cm)
Earth forms (left end)	4,500	0
Oldest mineral on Earth (zircon)	4,400	1
Oldest known rock	4,000	
Oldest fossils	3,500	
Oldest ocean floor	200	
Today (right end)	0	45

Material	Where	Age
Meteorite fragments (Canyon Diablo)	Arizona, USA	about 4.56 billion years
Moon rocks from the lunar highlands	Brought back by Apollo	up to about 4.4 billion years
Oldest Earth mineral (zircon)	Jack Hills, Australia	about 4.4 billion years
Oldest known Earth rock (Acasta Gneiss)	Canada	about 4.0 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years
Oldest ocean floor	Western Pacific and Atlantic margins	about 180–200 million years

Day 4: Why the Ocean Is Young and the Continents Old

D4-A Guided notes (fill in as your teacher goes)

Left page

1. Old ocean crust is made at a _____ and moves away about 2.5 cm a year (25 km per million years).
2. A piece of crust travels 3,000 km. Time = 3,000 ÷ 25 = _____ million years. In 200 million years it travels 200 × 25 = _____ km, so it reaches a trench and sinks. No ocean floor is older than about _____ million years.
3. Ocean crust has a density of about **3.0 g/cm³**. Continental crust is about **2.7 g/cm³**. The _____ plate sinks (*subduction*).
4. Each continent has an ancient core. Volcanic arcs and pieces of crust are added to its edge over time, like rings on a tree. The edge is _____ than the core.

D4-B Density table

Left page, for the sink-or-float lab

Block	Mass (g)	Volume (cm³)	Density (g/cm³) = mass ÷ volume	Sinks or floats?
Wood block (stand-in for a continent)				
Clay block (stand-in for an old, cold ocean plate)				

Water has a density of 1.0 g/cm³. Which block is denser? _____ What does the denser one do when they meet? _____

D4-C SketchNotes checklist

Right page, above your notes and drawing

Turn in a photo of your SketchNotes on Canvas: Lesson 4: SketchNotes.

Check	Half page of notes	Check	Cross-section drawing
☐	Where ocean crust is made	☐	Ridge and plate moving away
☐	How it moves and sinks	☐	Trench and volcanic arc
☐	How continents grow from a core	☐	Continent with an ancient core and a younger edge
☐	Density numbers	☐	Density labels and a carbon arrow

Day 5: Evidence for Moving Crust

D5-A Magnetic stripe bands

Left page, for the paper strip model

Color both strips with this table (2 cm = 1 million years). Start at the end you will push through first.

Band (from the end you push first)	Color	Length	Age it represents
1	Reversed (blue)	3.4 cm	about 3.6–5.3 million years
2	Normal (red)	2.0 cm	about 2.6–3.6 million years
3	Reversed (blue)	3.6 cm	about 0.8–2.6 million years
4	Normal (red)	1.6 cm	0–0.8 million years (youngest)

D5-B Hawaii and GPS

Right page

Volcano or island	Distance northwest of Kīlauea (km)	Age of its oldest rock (million years)
Kīlauea (Hawai'i Island, active now)	0	0
Kaua'i	about 500	about 5
Midway Atoll	about 2,500	about 28

GPS today: about 7 to 8 cm per year (Pacific plate near Hawaii). 1 km per million years = 0.1 cm per year.

Kīlauea to Kaua'i: 500 ÷ 5 = _____ km per million years = _____ cm per year.

Kīlauea to Midway: _____ ÷ _____ = _____ km per million years = _____ cm per year.

D5-C Fossil table

Right page, with your Pangaea puzzle

Organism	What it was	Found in
Mesosaurus	freshwater reptile	South America and southern Africa
Glossopteris	seed fern (heavy seeds)	South America, Africa, India, Australia, Antarctica
Lystrosaurus	land animal	Africa, India, Antarctica

D5-D Evaluate the evidence

Right page

Rate strength from 1 (weak) to 4 (very strong).

Line of evidence	Measured numbers?	Past or present motion?	Independent of the others?	Strength (1–4)
Rock ages across a ridge				
Magnetic stripes				
GPS and Hawaiian island ages				
Matching fossils and rocks				

Day 6: Evaluate the Theory, Tell Earth's Story

D6-E Three buckets

Left page, sort your evidence log rows

The seafloor is young because...	The continents are old because...	The early record is missing because...

D6-A Part A sentence frames

Left page

- **Claim:** Plate tectonics explains the ages of Earth's crustal rocks (very well / partly / poorly) because _____.
- **Ocean evidence:** The seafloor is youngest at the _____ and gets older with _____; the oldest ocean floor is only about _____ million years old.
- **Continent evidence:** Continental crust as old as _____ exists, and in North America the crust gets _____ toward the edges.
- **Reasoning:** Because ocean plates are _____ than continental crust, they _____ at subduction zones, while continents _____.
- **Limit or counterclaim:** One thing this evidence cannot show is _____. OR: Someone might say the ocean floor is young because _____. This does not fit the evidence because _____.

D6-D Peer feedback

Right page

Partner's name	One star (something strong)	One step (one thing to improve)

D6-B Evidence cards: cut apart, order oldest to newest, glue on a strip

Right page

Evidence card	What it shows
Meteorites: about 4.56 billion years	The solar system formed about 4.56 billion years ago.
Moon highland rocks: up to about 4.4 billion years; heavily cratered	Solid crust existed early; the early solar system had many impacts.
Earth's oldest zircons: about 4.4 billion years	Solid crust and minerals had formed on Earth by then.
Oldest Earth rock (Acasta Gneiss): about 4.0 billion years	Rock has survived for 4 billion years in the continents.
Oldest fossils (stromatolites): about 3.5 billion years	Life existed by then.
Oldest ocean floor: about 200 million years	Ocean crust is recycled; the early record is not on the ocean floor.

D6-C Part B story frames

Right page, under your evidence cards

- About _____ billion years ago, _____ happened. The evidence is _____.
- About _____ billion years ago, _____ happened. The evidence is _____.
- About _____ billion years ago, _____ happened. The evidence is _____.
- Earth's earliest record is missing because _____.

Day 7: Review Day

D7-A Self-rating chart

Left page

Rate each target from 1 (I need help) to 4 (I could teach it). Use your CFA scores to check your rating.

Target	My rating (1–4)	My CFA score	What I will do today
ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.			
ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.			
ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.			
ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.			
ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.			